

DOCUMENT CONTROL

Technical Architecture Brief

Engineering architecture brief with appendices imported from working papers and scan-heavy captures.

Document ID	TEC-001
Owner	Platform Engineering
Version	0.8 draft
Prepared Date	2024-02-29
Status	Engineering review draft
Classification	Internal / Technical

Synthetic corporate document for demo, evaluation, and retrieval workflows only. All parties, people, systems, repositories, and events are fictional.

1. Objective and Scope

This brief describes the architecture of Meridian's Knowledge Operations Platform, including the vendor-managed retrieval layer, internal policy services, content pipelines, and observability controls used for the Q2 2024 pilot.

- Scope includes retrieval path, orchestration, document ingestion, policy retrieval, and logging boundaries.

Technical Reference	Detail
TEC-REF-01	1. Objective and Scope — architecture note maintained by Platform Engineering.
TEC-CTRL-01	Control dependency reviewed alongside policy, contract, and evidence workflows.

2. System Context

The platform combines internal indexing pipelines with the Northwind managed retrieval cluster. User traffic enters through Meridian identity and gateway services before requests are brokered to retrieval and summarisation components.

- Primary dependencies: SSO, API gateway, Northwind cluster, internal policy service, event telemetry.

Technical Reference	Detail
TEC-REF-02	2. System Context — architecture note maintained by Platform Engineering.
TEC-CTRL-02	Control dependency reviewed alongside policy, contract, and evidence workflows.

3. Data Classification and Routing

Corporate data is classified at ingestion. Regulated or confidential content is routed to approved stores, and non-production experimentation uses synthetic or anonymised material only.

- Routing labels drive storage tier, retention policy, and retrieval permissions.

Technical Reference	Detail
TEC-REF-03	3. Data Classification and Routing — architecture note maintained by Platform Engineering.
TEC-CTRL-03	Control dependency reviewed alongside policy, contract, and evidence workflows.

4. Retrieval Pipeline

Documents are chunked, embedded, indexed, and routed through reranking and answer-generation layers. The design intentionally separates retrieval from generation to preserve evidence and auditability.

- Key controls: chunk metadata, repository lineage, grounded answer citations.

Technical Reference	Detail
TEC-REF-04	4. Retrieval Pipeline — architecture note maintained by Platform Engineering.
TEC-CTRL-04	Control dependency reviewed alongside policy, contract, and evidence workflows.

5. Security Controls

Authentication, authorisation, logging, and change review are enforced at the Meridian boundary. Vendor-managed components supply telemetry and admin-activity exports to Meridian security tooling.

- Quarterly privileged review and admin approval evidence are required for vendor support access.

Technical Reference	Detail
TEC-REF-05	5. Security Controls — architecture note maintained by Platform Engineering.
TEC-CTRL-05	Control dependency reviewed alongside policy, contract, and evidence workflows.

6. Availability and Recovery

The initial pilot targets operational resilience through fallback search paths and dual-run validation, while commercial SLA negotiations remain open.

- Current vendor SLA is 99.5% monthly with a temporary pilot-phase exception request in place.

Technical Reference	Detail
TEC-REF-06	6. Availability and Recovery — architecture note maintained by Platform Engineering.
TEC-CTRL-06	Control dependency reviewed alongside policy, contract, and evidence workflows.

7. Model and Evaluation Layer

Generation and retrieval settings are evaluated against internal groundedness, adherence, and latency criteria. Structured extraction tasks share the same evidence-bearing retrieval layer.

- Primary tasks include extraction, summary, checklist, contract risk extraction, and CV analysis.

Technical Reference	Detail
TEC-REF-07	7. Model and Evaluation Layer — architecture note maintained by Platform Engineering.
TEC-CTRL-07	Control dependency reviewed alongside policy, contract, and evidence workflows.

8. Open Risks

Open issues include vendor liability, transfer mechanisms for EU analytics data, and explicit incident-notification timing. These issues are tracked outside the architecture design but affect go-live gating.

- The technical design itself remains viable; go-live is conditional on non-technical remediation.

Technical Reference	Detail
TEC-REF-08	8. Open Risks — architecture note maintained by Platform Engineering.
TEC-CTRL-08	Control dependency reviewed alongside policy, contract, and evidence workflows.

9. Operational Readiness

Cutover requires validated ingestion, observability, dashboard alerts, on-call runbooks, and approved evidence routing. The pilot also depends on archiving technical annexes and diagrams into the document library.

- Pages 12-15 intentionally preserve scan-heavy working annexes to test OCR-routing behavior.

Technical Reference	Detail
TEC-REF-09	9. Operational Readiness — architecture note maintained by Platform Engineering.
TEC-CTRL-09	Control dependency reviewed alongside policy, contract, and evidence workflows.

10. Appendix Index

Appendices include network capture diagrams, control mapping notes, and engineering whiteboard snapshots. Several pages are image-only imports from working papers and may require OCR fallback.

- The frontend demo intentionally models these pages as difficult for OCR and indexing.

Technical Reference	Detail
TEC-REF-10	10. Appendix Index — architecture note maintained by Platform Engineering.
TEC-CTRL-10	Control dependency reviewed alongside policy, contract, and evidence workflows.

11. Change Log

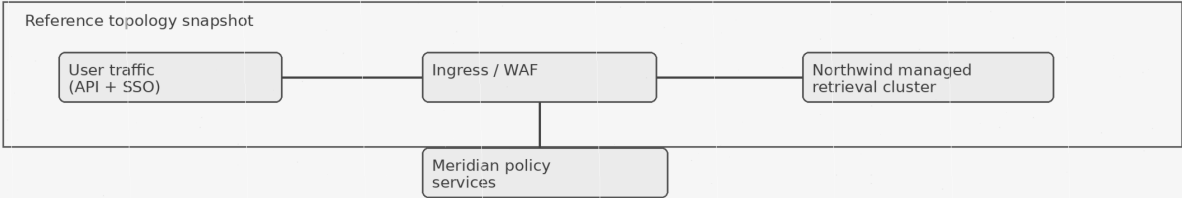
Draft 0.8 integrates vendor-review feedback, adds policy-service boundaries, and records the need to preserve evidence-bearing document lineage in downstream workflows.

- Next revision expected after the procurement and privacy gates are cleared.

Technical Reference	Detail
TEC-REF-11	11. Change Log — architecture note maintained by Platform Engineering.
TEC-CTRL-11	Control dependency reviewed alongside policy, contract, and evidence workflows.

Appendix A — topology capture

Appendix capture — imported from engineering working papers



- 01. Northwind retrieval nodes and Meridian gateway dependency map.
- 02. Note handwritten arrows showing fallback path and manual failover conditions.
- 03. Included to test OCR routing and diagram handling.

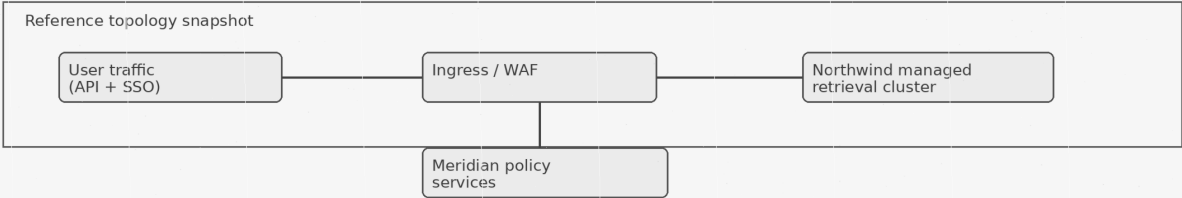
Appendix B — control mapping snapshot

Appendix capture — imported from engineering working papers

- 01. Annotated list of control IDs against platform components.
- 02. Contains mixed handwritten and typed notes copied from workshop materials.
- 03. Text density is intentionally uneven.

Appendix C — whiteboard incident flow

Appendix capture — imported from engineering working papers



- 01. Escalation path from service alert to security and legal review.
- 02. Critical timings under discussion: 24h, 48h, and vendor 72h counterproposal.
- 03. Used to simulate OCR-hard engineering notes.

Appendix D — deployment checklist capture

Appendix capture — imported from engineering working papers

- 01. Pilot launch readiness notes, owner initials, and references to evidence artifacts.
- 02. Contains small-font line items and embedded identifiers.
- 03. Image-only page for parser stress testing.